

Zhengxu (Jason) Yan

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EDUCATION

Stanford University <i>Master of Science in Computer Science</i>	Expected Graduation: April 2027 GPA: 4.15/4.00
University of California, Berkeley <i>Bachelor of Arts in Computer Science and Data Science</i>	August 2021 - May 2025 GPA: 3.95/4.00, High Distinction
• <i>Selected Coursework: Optimization Models (EECS 127), Probability for Data Science (Data C140), Computer Security (CS 161), Operating Systems (CS 162), Efficient Algorithm (CS 170), Computer Vision (CS 180), Artificial Intelligence (CS 188), Machine Learning (CS 189), LLM Agents (CS 194-196).</i>	

PUBLICATIONS

- Ye, Z., **Yan, Z.**, He, J., Kasriel, T., Yang, K., & Song, D. VERINA: Benchmarking Verifiable Code Generation. Accepted at *International Conference on Learning Representations (ICLR 2026)*. <https://arxiv.org/abs/2505.23135>
- Schaumann, D., **Yan, Z.**, Haiman, E., & Kalay, Y. E. A Deep Reinforcement Learning Framework for Multi-Dimensional Adaptability in Building Operations. Under review at *Journal of Building Engineering*.
- Luo, M., Hao, A., **Yan, Z.**, Cao, C., & Nguyen, Q. L. N. DiT-Serve: An Efficient Serving Engine for Diffusion Transformers. Under review at *ICML 2026*.
- **Yan, Z.**, Li, H., & Feng, Y. NoveltyRank: Estimating Conceptual Novelty of AI Papers. <https://arxiv.org/abs/2512.14738>
- **Yan, Z.**, Dube, V., Heselton, J., Johnson, K., Yan, C., Jones, V., Blaskewicz Boron, J., & Shade, M. (2024). Understanding older people's voice interactions with smart voice assistants: a new modified rule-based natural language processing model with human input. *Frontiers in digital health*, 6, 1329910. <https://doi.org/10.3389/fdgth.2024.1329910>
- Jones, V. K., Yan, C., Shade, M. Y., Boron, J. B., **Yan, Z.**, Heselton, H. J., Johnson, K., & Dube, V. (2024). Reducing Loneliness and Improving Social Support among Older Adults through Different Modalities of Personal Voice Assistants. *Geriatrics (Basel, Switzerland)*, 9(2), 22. <https://doi.org/10.3390/geriatrics9020022>

WORK EXPERIENCE

Graduate Research Assistant (Supervised by Dr. Chealsea Finn) <i>Intelligence through Robotic Interaction at Scale Lab (IRIS), Stanford AI Lab (SAIL)</i>	November 2025 - Present Stanford, CA
• Conducting research on scaling Feedback Descent for iterative model improvement. • Investigating structured critique models and multi-round refinement pipelines for research plan generation and other tasks. • Aiming to develop a scalable feedback framework that improves performance without task-specific retraining.	
Graduate Research Assistant (Supervised by Dr. Diyi Yang and Dr. Alexander Spangher) <i>Stanford Social and Language Technologies Lab (SALT), Stanford AI Lab (SAIL)</i>	November 2025 - Present Stanford, CA
• Building large-scale pipelines to extract structured metadata and dataset claims from recent NLP papers. • Developing representations for dataset novelty, adoption, and conceptual contribution. • Working toward empirical analyses of dataset trends and impact across major NLP venues.	
Research Assistant (Supervised by Dr. Ion Stoica) <i>UC Berkeley Sky Computing Lab</i>	February 2024 - October 2025 Berkeley, CA
• Competitively selected through UC Berkeley EECS Diversifying Access to Research in Engineering (DARE) and Sky Summer Undergraduate Programs to join the DiT-Serving Project as a research assistant. • Led a team of 5 research assistants in integrating Ring Attention and Brick Attention into Diffusion Transformer (DiT) models, pioneering scalable video generation techniques. • Directed continuous batching strategies to optimize system throughput, enhancing performance efficiency across video processing requests.	
Research Assistant (Supervised by Dr. Yehuda Kalay and Dr. David Schaumann) <i>UC Berkeley College of Environmental Design</i>	February 2024 - September 2025 Berkeley, CA
• Competitively selected to join the Smart Hospital Project through UC Berkeley Undergraduate Research Apprenticeship Program. • Led a team of 3 research assistants in the development and implementation of a deep reinforcement learning-based smart building management system for the Cardiac Catheterization Lab at St. Bernardine Medical Center.	
Machine Learning (ML) Engineer Lead (Supervised by Dr. Stefano Bertozzi) <i>UC Berkeley School of Public Health</i>	January 2024 - September 2025 Berkeley, CA
• Selected as lead computer scientist for Rapid Reviews: Infectious Diseases via the UC Berkeley CDSS Discovery Program.	

- Independently leveraged a fine-tuned Large Language Model (LLM) to efficiently categorize and identify preprints within the RR\ID domain, significantly enhancing peer review efficiency.
- Led a team of 3 UC Berkeley CS students in utilizing LLM APIs to analyze and provide insights on medical preprints, supporting the peer review and decision-making process.
- Directed a team of 5 UC Berkeley CS students in engineering scripts to automate the collection of medical preprints from multiple servers, streamlining data acquisition and analysis workflows.
- Collaborated with the Dean of the UC Berkeley School of Public Health and researchers from UCSF to develop automated systems for the academic review of infectious disease papers, set to be published by MIT Press.

Teaching Assistant for Large Language Model Agents (CS 194-196) (Supervised by Dr. Dawn Song) August 2024 - May 2025
Department of Electrical Engineering and Computer Sciences at UC Berkeley *Berkeley, CA*

- Coordinated course logistics, managed lecture recordings, enhanced assignment materials, and provided academic support by addressing student inquiries.
- Assisted in organizing and coordinating the LLM Agents Hackathon, hosted by Berkeley RDI in conjunction with the LLM Agents MOOC, designed to foster innovation, expand the AI agent community, and advance LLM agent technology.

Research Assistant (Supervised by Dr. Valerie Jones) April 2023 – September 2023
University of Nebraska *Lincoln, NE*

- Collaborated with gerontology and nursing researchers to develop a natural language processing (NLP) model designed to automate the processing of speech-to-text data from user interactions with AI-enabled smart voice assistants.

Backend Engineer Lead October 2022 – December 2022
Coffee Tea, Inc. *Berkeley, CA*

- Led a team of 3 UC Berkeley computer science students in backend development for a social platform aimed at broadening access to the college application process by facilitating coffee chat connections between applicants and current college students.
- Directed the design and development of a comprehensive suite of backend REST APIs using FastAPI, Poetry, Alembic, and PostgreSQL, covering modules such as authentication, profile management, recommendation systems, and payment processing.

PROJECTS

Machine Learning

Multi-Agent LLM Trading System | *Python, Pytorch, Tensorflow, AutoGen* August 2024 – December 2024

- Developed a multi-agent trading system integrating LLMs with specialized agents for multimodal data processing, layered memory, and Retrieval-Augmented Generation, aiming to exceed standard prediction models by 5-7% in market accuracy.

IM2SPAIN Project | *Python* January 2024 – May 2024

- Employed nearest neighbors (k-NN) to predict geographic coordinates based on CLIP embeddings of geo-tagged images from Flickr, capturing diverse landscape features across Spain.

MNIST Competition | *Python, Pytorch* January 2024 – May 2024

- Engaged in the Kaggle MNIST classification challenge, leveraging a diverse array of machine learning techniques including Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Logistic Regression, Multi-layer Perceptron (MLP), Support Vector Machine (SVM), and Convolutional Neural Networks (CNN) to optimize prediction accuracy.

Language Identification RNN | *Python, Tensorflow* August 2023 – December 2023

- Engineered an RNN to identify word languages, dynamically processing variable-length character inputs to achieve over 81% test set accuracy.

Graph Partitioning for Unsupervised Learning | *Golang, AWS, Google Cloud* August 2022 – December 2022

- Developed graph partitioning algorithms to perform unsupervised machine learning tasks, using the Kernighan-Lin algorithm.
- Implemented a relational database and Golang backend on AWS and Google Cloud platforms to manage outputs and facilitate the running of algorithms.

Systems Programming and Software Development

Pintos Operating System | *C, Rust* January 2023 – May 2023

- Led the development of a comprehensive Pintos operating system, focusing on systems programming, memory allocation, resource management, file systems, networking, and security, enhancing system functionality and efficiency.

Secure Client Application | *Golang* January 2023 – May 2023

- Developed a secure client Golang application incorporating cryptographic primitives to manage authentication, file management, sharing, and access revocation, significantly enhancing data security and user privacy.

Gitlet Version Control System | *Java* January 2021 – May 2022

- Created “Gitlet,” a Git-like Version Control System, to streamline tracking and management of code changes across projects.

TECHNICAL SKILLS

Areas: Machine Learning, Deep Reinforcement Learning, Operating Systems, Natural Language Processing, Computer Vision

Languages: Python, Java, C/C++, Golang, SQL (Postgres), JavaScript, Rust, Bash, HTML/CSS

Developer Tools: Git, AutoGen, Stable Baselines, Docker, FastAPI, Django, Google Cloud Platform, AWS, Vercel

Libraries: TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, CV2, Scrapy, OpenAI Gym